

CM-GAN: A Cross-Modal Generative Adversarial Network for Imputing Completely Missing Data in Digital Industry

Mingyu Kang^{1b}, Ran Zhu^{1b}, Duxin Chen^{1b}, *Member, IEEE*, Xiaolu Liu^{1b}, and Wenwu Yu^{1b}, *Senior Member, IEEE*

Abstract—Multimodal data fusion analysis is essential to model the uncertainty of environment awareness in digital industry. However, due to communication failure and cyberattack, the sampled time-series data often have the issue of data missing. In some extreme cases, part of units are unobservable for a long time, which results in complete data missing (CDM). To impute missing data, many models have been proposed. However, they cannot address the CDM issue, because no observation data of the unobservable units can be obtained in this case. Thus, to address the CDM issue, a novel cross-modal generative adversarial network (CM-GAN) is proposed in this article. It combines the cross-modal data fusion technique and the deep adversarial generation technique to construct a cross-modal data generator. This generator can generate long-term time-series data from widely existing spatio-temporal modal data in modern industrial system, and then impute missing value by replacing them with generated data. To test the performance of CM-GAN, extensive experiments are conducted on photovoltaic (PV) power output dataset. Compared with other baseline models, the performance of CM-GAN is generally better and reaches the state-of-the-art level. Moreover, sufficient ablation studies are conducted to present the contribution of the cross-modal data fusion technique and show the reasonability of parameter settings of CM-GAN. Apart from this, some prediction experiments are also conducted. The results show that the PV data recovered by CM-GAN can provide more predictability information for improving the prediction accuracy of deep learning model.

Index Terms—Complete data missing (CDM), cross-modal data fusion, digital industry, generative adversarial network (GAN), predictability.

I. INTRODUCTION

WITH the rapid development of sensor network, multimodal data fusion analysis has become more essential to model the uncertainty of environment awareness in modern digital industry [1], [2]. Many multimodal learning theories are proposed recently and applied in various industrial scenarios based on neural network [3], [4], [5]. They often require large-scaled and high-quality data to train the designed models. However, in practical industrial scenarios (e.g., photovoltaic (PV) power system), the data collected from the sensor network often have the issue of data missing and outliers produced in communication failure [6], [7] and cyberattack [8], [9], as shown in Fig. 1. Moreover, the data cannot always be collected in some cases, e.g., the PV units is newly built [10]. That is, the issue of complete data missing (CDM) occurs, which means part of PV units are unobservable in the view of system operator for a long time, and no data of the newly built PV units can be used for analysis.

To impute the missing data in modern digital industry, the tensor-based methods were introduced to reconstruct the missing value from a spatio-temporal tensor based on the principle of low-rank tensor factorization. By modeling the time-series as a multiway tensor, Zhao et al. [11] proposed a low-CANDECOMP/PARAFAC-rank tensor factorization method to capture the global and local interaction information in the tensor, and then infer the missing data and outliers by a fully Bayesian treatment. To capture the periodic pattern in tensor, Chen et al. [12] proposed a Bayesian Gaussian CANDECOMP/PARAFAC (BGCP) method which additionally introduced conjugated Gaussian–Wishart priors. To capture the strong local consistency in tensor, Chen et al. [12] also proposed a low-rank autoregressive tensor completion (LATC) method by introducing temporal variation as a regularization term.

Recently, some deep learning methods were proposed to impute and predict time-series in digital industrial system. Ko et al. [13] proposed a stackable long short-term memory neural network with multilevel residual structure to predict time-series in power grid operations. To improve the computational efficiency and prediction accuracy, Xia et al. [14] proposed a stackable gate recurrent unit neural network with

Manuscript received 28 September 2022; revised 25 December 2022 and 10 April 2023; accepted 30 May 2023. Date of publication 23 June 2023; date of current version 1 March 2024. This work was supported in part by the National Key Research and Development Program of China under Grant 2022ZD0120004; in part by the National Natural Science Foundation of China under Grant 62233004, Grant 62273090, Grant 62073076, and Grant 62203210; in part by the Jiangsu Provincial Key Laboratory of Networked Collective Intelligence under Grant BM2017002; in part by the Scientific Research Foundation for the Introduction of Talents of NJIT under Grant YKJ202015; in part by the Jiangsu Postdoctoral Science Foundation under Grant 2021K397C; and in part by the Natural Science Foundation of Jiangsu Province of China under Grant BK20210929. (Corresponding authors: Duxin Chen; Wenwu Yu.)

Mingyu Kang and Ran Zhu are with the School of Cyber Science and Engineering, Southeast University, Nanjing 211189, China (e-mail: kangmingyu.china@gmail.com; gemina_cat@163.com).

Duxin Chen is with the School of Mathematics, Southeast University, Nanjing 211189, China (e-mail: chendx@seu.edu.cn).

Xiaolu Liu is with the School of Automation, Nanjing Institute of Technology, Nanjing 211167, China (e-mail: emspc@126.com).

Wenwu Yu is with the Frontiers Science Center for Mobile Information Communication and Security, School of Mathematics, Southeast University, Nanjing 210096, China, and also with the Purple Mountain Laboratories, Nanjing 211102, China (e-mail: wwyu@seu.edu.cn).

Digital Object Identifier 10.1109/TNNLS.2023.3284666

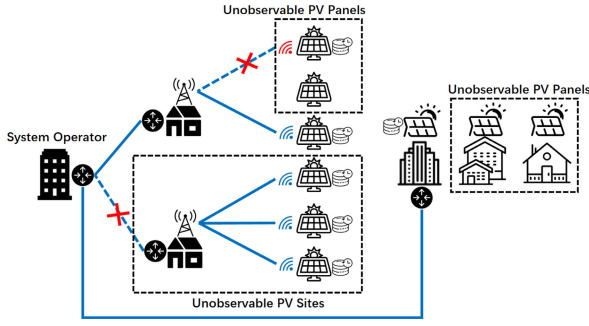


Fig. 1. Digital industrial system with unobservable PV units.

improved self-feedback connections. Wang et al. [15] proposed a transformer-based method based on multihead attention mechanism. The method adopted an one-encoder multi-decoder structure to realize joint prediction of multienergy loads.

Moreover, generative models were also introduced to model the uncertainty of industrial system, and the missing value can be imputed by resampling from the learned data distribution. Chen et al. [16] proposed a generative adversarial network (GAN)-based method to generate renewable energy scenarios' data via unsupervised way. The trained GAN can generate a group of time-series trajectories that cover the real trajectories. To impute data with high missing rate, Zhang et al. [17] proposed a SolarGAN model to impute the missing renewables' data when the missing rate was up to 90%. Qiao et al. [18] proposed the method for data generation of renewable power using stable and controllable GAN with transparent latent space, which improves the training stability of the previous GAN models. To capture the complex dynamics of the weather, Liang and Tang [19] proposed a distribution-free approach for wind power scenario data generation. It uses GAN combined with the reinforcement learning technology and has an advantage of avoiding manual labeling. Yuan et al. [20] proposed a C-StyleGAN2-SE model to achieve the spatio-temporal dynamics characteristics of renewable energy power system. C-StyleGAN2-SE can be used to control and mix the styles of generated scenarios by meteorological information serving as conditional inputs.

Some methods were proposed to estimate the missing value via multiview way [21], [22]. They tackle the task of learning latent representation from multiple views of data, and then impute missing views. To eliminate the heterogeneous gaps across different views, Peng et al. [23] proposed a view-consensus constraint to project raw data into a feature space and make these features embrace geometric consistency. Lin et al. [24] also proposed a contrastive learning framework to impute missing views by maximizing the mutual information across different views. Yao et al. [25] proposed a convolution-based attention mechanism to capture the time-series characteristics from the spatial and temporal views.

To further make full use of multimodal data, some methods were proposed to recover the missing data via cross-modal way, which translates one or many modals into the missing modal [2]. Zeng et al. [26] proposed a cross-modal method to capture the interactive relationships among

multiple objects from visual and textual contents, and then realize video moment retrieval using the relationships to identify the differences among similar video moment candidates. Zhang et al. [27] proposed an aggregation-based graph convolutional hashing model to learn unified binary codes across multiple modals. To capture fine-grained correlations across different modals, Zhao et al. [28] proposed a dynamic cross-modal attention network (CMAN) for relation extraction of joint entity. Note that there are two main differences between the multiview method and cross-modal method, which are listed as follows. 1) Multiview method is designed for data collected from multiple views of only one object, but cross-modal method is for multiple objects. 2) Multiview method focuses on latent representation learning, but cross-modal method pays more attention on data fusion between different modal data.

Although such imputation methods have been proposed, they cannot address the CDM issue because of zero-shot learning property, that is, no observation data can be obtained for the analysis of missing data because part of units are completely unobservable. Thus, the missing time-series data can only be recovered from the other modals via an unsupervised way. To address this issue, a cross-modal GAN (CM-GAN) is proposed to overcome the low imputation accuracy in the CDM case. In CM-GAN, a cross-modal data fusion module is designed to fuse multiple modal data into a concatenated feature vector, and GAN is used to translate the feature vector into the completely missing time-series via an unsupervised way. To guide the generation process of GAN by additional modal information, the conditional Wasserstein GAN is used to be the basic framework of CM-GAN. The Wasserstein measurement is used to train the GAN more stably.

Thus, the main contributions of this article are as follows.

- 1) CM-GAN is proposed to address the CDM issue in modern digital industry. In CM-GAN, a cross-modal data fusion technique is specially designed to impute completely missing data from multimodal data.
- 2) Lots of experiments are conducted to test the performances of CM-GAN on PV power dataset, which is a typical application of modern digital industry. The results show that the performance of CM-GAN is generally better than other baseline models and reaches the state-of-the-art level. Moreover, sufficient ablation studies are also conducted to verify the contribution of the cross-modal data fusion technique and guide the settings of CM-GAN.
- 3) Moreover, sufficient prediction experiments are also conducted. The results show that the data generated by CM-GAN provide more predictability information for improving prediction accuracy. This enables the commonly used prediction models to predict long-term time-series more accurately in industrial system.

The rest of this article is organized as follows. Section II presents the principle of CM-GAN and the implements of data generation. Section III discusses the experimental design and the results of assessment experiments. Section IV concludes this article.

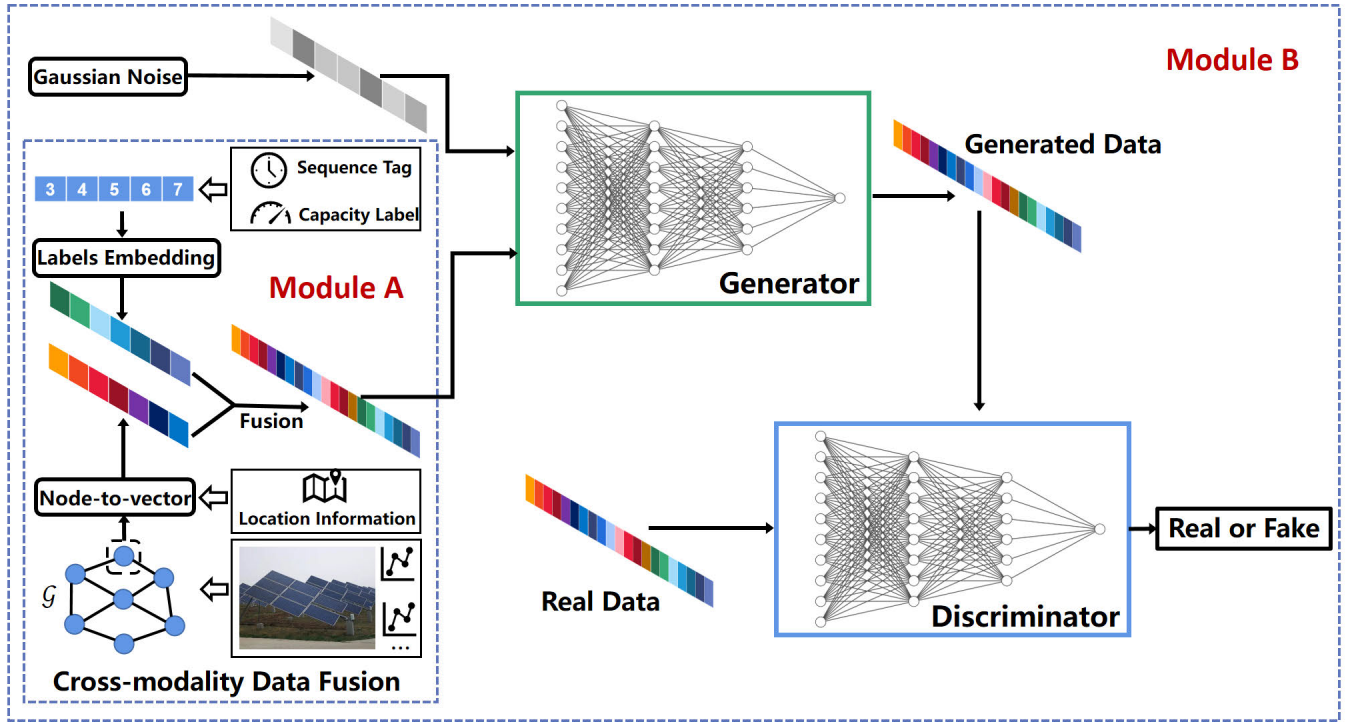


Fig. 2. Schematic of the CM-GAN architecture.

II. METHODOLOGY

To address CDM problem in an industrial system, e.g., PV power system, CM-GAN is proposed to impute complete missing time-series data from the other modal data, including GPS data and capacity labels. The GPS data induce a weighted graph $\mathcal{G} = (V, E)$. In graph $\mathcal{G}$, V represents the node set of all units and E represents the set of edges. Meanwhile, the capacity is a numerical label denoted as C . Then, this problem can be considered as the estimation of probability distribution

$$P_r = P\left(\{X_j\}_{j=1}^m | \{X_i\}_{i=1}^N \setminus \{X_j\}_{j=1}^m, \mathcal{G}, C\right) \quad (1)$$

where $\{X_i\}_{i=1}^N$ is the set of all units, and $\{X_j\}_{j=1}^m$ is the set of unobservable units, respectively. Then the unobservability rate can be defined as

$$\eta = \frac{m}{N} \quad (2)$$

which represents the ratio of the number of unobservable units to all units. One purpose of the proposed CM-GAN model is to fit this distribution. However, due to the heterogeneous representation of various model data, these data need to be transformed into isomorphic vectors as the input of the GAN model before model training. Therefore, the CM-GAN is mainly designed as two modules, as shown in Fig. 2. The different modal data are first transformed into isomorphic vectors after cross-modal data fusion, and then as the input of GAN with noise sampling. Finally, the time-series data of the unobservable units are generated by the trained generator. The process constructs a cross-modal data generation pipeline.

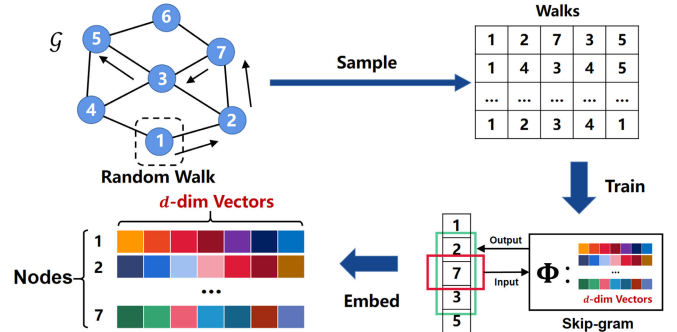


Fig. 3. Flowchart of the node-to-vector algorithm.

A. Cross-Modal Data Fusion

To make full use of multiple modal data and avoid the loss of model information, a node-to-vector algorithm [29], a graph embedding method, is adopted in this article. It is used to transform various graphs into vectors. In this article, this algorithm is applied to embed the location information of each unit, which are represented as longitude and latitude. Before graph embedding, the location data are preprocessed to be the weighted graph $\mathcal{G} = (V, E)$. For different nodes i and j , the weight W_{ij} of the edge is calculated by the spherical distance between their longitudes and latitudes.

For the obtained graph $\mathcal{G}$, a map $\Phi: V \rightarrow \mathbb{R}^d$ is found to extract structural features from adjacency, and then transforms these features into d -dimensional (d -dim) vectors, where d is the dimension of the latent space. The flowchart of this process is shown in Fig. 3. As can be seen, the algorithm first samples the sequential stream by deep random walk with the rollback-allowed strategy, which combines both the deep-first search and breadth-first search. Then, these stream samples are input

to the skip-gram model [30] to train parameters. Note that the expected d -dim vectors are the parameters of the skip-gram model. When the parameters are trained, the d -dim vectors record the correlation map Φ from each input node to the corresponding neighbors. Finally, the vectors of each node are achieved, and the dot product of two d -dim vectors records the structural similarity of the corresponding nodes.

Apart from these data that can be represented as graphs, some other modals, including sequence tag and capacity label, are collected as discrete labels. Among them, the sequence tag is used to explicitly encode the correlation of time-series data at different time steps. The capacity label records the upper bound of power output of a unit. To vectorize these labels, the embedding layer technology is introduced to raise the representation dimension of them. In the practical operation of the embedding layer, a matrix is used to transform the labels into a higher dimension latent space. The parameters of this matrix are shared with each label and trained by the backpropagation algorithm.

Finally, diverse modal data are all vectorized and concatenated to be the fusion vectors. These multimodal fusion vectors are used as the input of downstream module, namely, the GAN module, as shown in Fig. 2.

B. Generative Adversarial Network

GAN is a type of generative model, based on the deep learning architecture. It consists of two main components, a generator G and a discriminator D , which are two deep neural networks applied on different optimization tasks. It well addresses the intractable probabilistic computation problems by adversarial training methods. During the training process, the generator first tries its best to generate fake samples to fool the discriminator. And then, the discriminator receives two groups of samples, containing both real and fake labels. It also tries its best to discriminate them as real or fake. This constitutes a dynamic game process, which gradually reaches the Nash equilibrium by alternating iterative optimization. Finally, P_G is close to P_r , where P_G is the model distribution of the generator and P_r is described as (1), respectively.

Based on the principle of the conditional Wasserstein GAN [31], [32], CM-GAN is proposed. The schematic of CM-GAN is displayed in Fig. 2. Different from the traditional GAN models, it uses extra multimodal fusion data as conditional input, not only noise samples, as shown in Fig. 2. With enough training samples, the parameterized GAN can translate the shared distribution of various modal fusion data into the distribution of the time-series data by a cross-modal way. The fusion vectors are generated by the aforementioned methods, while the noise inputs are sampled for each iteration.

Besides, the model training process of CM-GAN is similar to the traditional GAN. But considering the property of the time-series data, the reconstruction error term is introduced into the optimization function of our model, which is added on the optimization object of the generator and ensures the convergence of model parameters. It can be represented as

$$L_{rc} = \|\tilde{x} - x\|^2 \quad (3)$$

where $\|\cdot\|$ means two-norm. Moreover, the x is the real time-series data, and $\tilde{x} \sim P_G$ is the estimated value. Then, the object functions can be defined as

$$\begin{aligned} L[G] &= -D(\tilde{x}) + \beta L_{rc} \\ L[D] &= D(\tilde{x}) - D(x) + \mu \left(\|\nabla_{\hat{x}} D(\hat{x})\|^2 - 1 \right)^2 \end{aligned} \quad (4)$$

where $(\|\nabla_{\hat{x}} D(\hat{x})\|^2 - 1)^2$ is the gradient penalty, which guarantees the Lipschitz continuity of the discriminant function. $\hat{x}$ is the result of linear random interpolation, equaling to $\epsilon x + (1 - \epsilon)\tilde{x}$. Here, $\epsilon \sim U(0, 1)$, which is a random variable conforming to 0–1 uniform distribution. Then, the discriminator is parameterized as $D_\omega(x)$ and the generator is parameterized as $G_\theta(z, v, t, c)$, respectively. Here, v is the multimodal fusion vector and z is the noise sample submitting to independent and identically distributed (i.i.d) Gaussian distribution $P(z) \sim N(0, \alpha)$ with α controlling the diversity of generated samples. Meanwhile, t is the sequence tag of time, and c is the capacity of each PV unit.

Then, the generator and discriminator minimize their own object functions, respectively, in the dynamic game process. Finally, P_G will be close to the real probability distribution of real samples.

C. Cross-Modal Time-Series Generation

In Sections II-A and II-B, the two main components of CM-GAN have been presented in detail. Then, the interactive process of multimodal information between these two components is described. Overall, there are two steps in the process of cross-modal time-series adversarial generation. It first vectorizes the multimodal information of each node and concatenates them into fusion vectors. Then, it generates training set and test set with the above vectors as input, and the corresponding time-series data are output. The CM-GAN model is trained by the generated training set. When the model parameters are converged, the generator can be used to impute time-series value of designated nodes. The detailed algorithm is summarized in **Algorithm 1**.

III. CASE STUDY

A. Datasets

The model performances will be tested on two different datasets, which are collected from the publicly available solar power dataset from National Renewable Energy Laboratory (NREL) [33]. The two datasets contain the time-series PV power output data of the utility-scale PV units, with 5-min time resolution, located in North Carolina (NC) and Arizona (AZ) in 2006, respectively. The number of PV units in the NC and AZ datasets is 149 and 49, respectively, while the installed capacity ranges from 20 to 39 MW and 39 to 118 MW, respectively. The PV site location distributions of the datasets are shown in Fig. 4.

B. Evaluation Indices

To evaluate the imputation performance of trained model from different perspectives, two main indices are selected,

Algorithm 1 Cross-Modal Time-Series Generation

Input: GPS data, capacity labels, time-series data,
batch size N , learning rate r , iterations m , n
Output: Converged generator of CM-GAN
/* **Step 1:** Cross-modal data fusion */
1 Generate $\mathcal{G}$ from GPS data;
2 walks $\leftarrow$ RandomWalk($\mathcal{G}$);
3 vectors $\leftarrow$ skip-gram(walks);
4 Get vectors v by concatenated vectors. Then, generate
training set and test set;
/* **Step 2:** GAN model training */
5 **while** not converged **do**
6 **for** iter = 1 to m **do**
7 Sample minibatch of $z \sim P(z)$, and $x \sim P_r(x)$
 with vectors v , sequence t and capacity c ;
8 Sample a random number $\epsilon \sim U(0, 1)$;
9 $\tilde{x} \leftarrow G_\theta(z, v, t, c)$;
10 $\hat{x} \leftarrow \epsilon x + (1 - \epsilon)\tilde{x}$;
11 Update ω by RMSprop optimizer with r :
 $\nabla_\omega \frac{1}{N} \sum_{i=1}^N D_\omega(\tilde{x}) - D_\omega(x) + \mu(\|\nabla_{\tilde{x}} D(\hat{x})\|^2 - 1)^2$;
12 **for** iter = 1 to n **do**
13 Sample minibatch of $z \sim P(z)$ and $x \sim P_r(x)$
 with vectors v , sequence t , and capacity c ;
14 $\tilde{x} \leftarrow G_\theta(z, v, t, c)$;
15 Update θ by RMSprop optimizer with r :
 $\nabla_\theta \frac{1}{N} \sum_{i=1}^N -D_\omega(\tilde{x}) + \beta \|\tilde{x} - x\|^2$;

namely, root mean square error (RMSE) and mean absolute error (MAE). They are defined as

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (x_i - \hat{x}_i)^2}$$

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |x_i - \hat{x}_i| \quad (5)$$

where x_i is the observed PV power output value, and $\hat{x}_i$ is the imputed value. Before calculating the values of indices, we normalize PV power output value into range from 0 to 1.

Moreover, to compare the performance improvement between two models M, M' on one index, the improvement can be defined as

$$\text{Improvement} = \frac{\text{Index}_M - \text{Index}_{M'}}{\text{Index}_{M'}} \times 100\%. \quad (6)$$

C. Experimental Settings

In our practical experiments, the generator and discriminator are constructed by two fully connected neural networks, respectively, as shown in Fig. 5. In the generator, the neural units in each hidden layer are [1024, 1024, 1024]. And the discriminator's are [512, 512, 512]. Besides, the activation

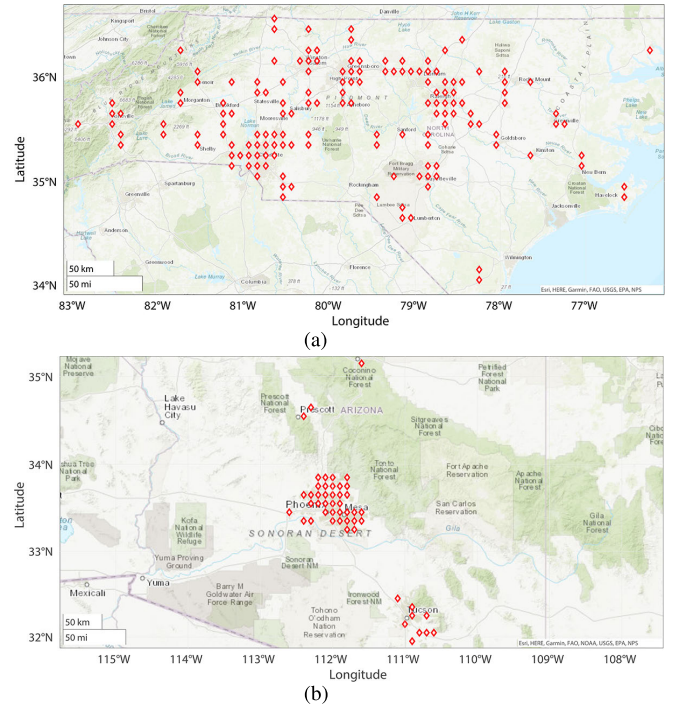


Fig. 4. Location distribution of PV units. (a) Locations of PV units in the NC dataset. (b) Locations of PV units in the AZ dataset.

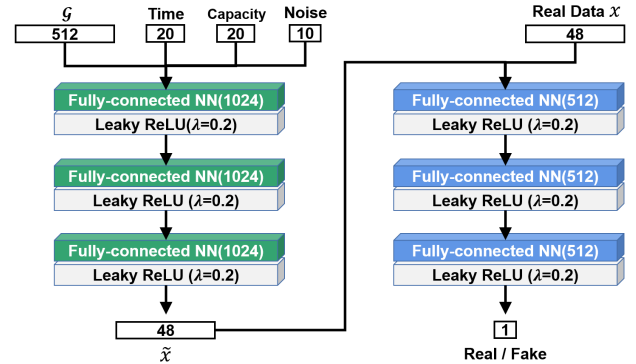


Fig. 5. Settings of the generator and discriminator.

function of each layer is the Leaky ReLU, whose slope parameter $\lambda = 0.2$. Graph $\mathcal{G}$ is embedded into 512-dim vectors. Moreover, some other labels, such as the sequence tags of time and the capacity of PV units, are embedded into 20-dim vectors. The noise is set as a 10-dim vector, which is sampled from standard normal distribution with $\alpha = 1$. The output dimension of the generator is set to be 48. The output of the discriminator is a single label, which is a 1-dim continuous value.

To train CM-GAN by **Algorithm 1**, the values of β and μ are set by $\beta = 10$ and $\mu = 10$, respectively. Moreover, the RMSprop optimizer is used to update parameters with ascending gradients. The RMSprop optimizer is initialized with learning rate $r = 0.00001$. Besides, the batch size N is set as 16. The iterations m and n are set as 5 and 3, respectively.

Extensive experiments are conducted to compare the performance of CM-GAN with that of the other baseline models. The experimental settings of these models are all presented as follows.

TABLE I
IMPUTATION PERFORMANCE OF EACH MODEL ON THE NC DATASET

η	5%		10%		15%		20%		25%	
	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE
BGCP	0.104	0.079	0.106	0.082	0.110	0.082	0.113	0.080	0.113	0.083
LATC	0.182	0.105	0.188	0.108	0.192	0.110	0.198	0.116	0.207	0.123
Transformer	0.118	0.088	0.121	0.090	0.125	0.092	0.127	0.093	0.130	0.094
MVSTGN	0.111	0.083	0.112	0.083	0.114	0.085	0.114	0.086	0.117	0.088
CMAN	0.112	0.081	0.116	0.085	0.119	0.088	0.125	0.089	0.129	0.095
SolarGAN	0.111	0.084	0.111	0.084	0.113	0.085	0.122	0.090	0.130	0.098
C-StyleGAN2-SE	0.104	0.076	0.105	0.077	0.109	0.081	0.110	0.082	0.111	0.083
CM-GAN	0.103	0.075	0.103	0.076	0.101	0.075	0.103	0.076	0.104	0.076
CM-GAN-G-C	0.261 (-60.5%)	0.226 (-66.8%)	0.261 (-60.5%)	0.227 (-66.5%)	0.261 (-61.3%)	0.227 (-67.0%)	0.263 (-60.8%)	0.227 (-66.5%)	0.263 (-60.5%)	0.228 (-66.7%)
CM-GAN-T	0.129 (-20.2%)	0.092 (-18.5%)	0.129 (-20.2%)	0.095 (-20.0%)	0.131 (-22.9%)	0.098 (-23.5%)	0.132 (-22.0%)	0.098 (-22.5%)	0.132 (-21.2%)	0.098 (-22.4%)
CM-GAN-T-G	0.105 (-1.9%)	0.077 (-2.6%)	0.105 (-1.9%)	0.077 (-1.3%)	0.106 (-4.7%)	0.078 (-3.8%)	0.107 (-3.7%)	0.078 (-2.6%)	0.107 (-2.8%)	0.078 (-2.6%)
CM-GAN-T-C	0.109 (-5.5%)	0.079 (-5.1%)	0.110 (-6.4%)	0.081 (-6.2%)	0.110 (-8.2%)	0.081 (-7.4%)	0.110 (-6.4%)	0.082 (-7.3%)	0.112 (-7.1%)	0.083 (-8.4%)

BGCP [12] and LATC [34] are tensor-based models. The number of rank in the two methods is both set as 10, and the other parameters are the same as the original papers. In the experiments, the data of the unobservable PV units are discarded as the missing data, and the BGCP and LATC are used to reconstruct the missing data from the reserved data tensor.

Moreover, some deep learning models are also selected, including Transformer [15], MVSTGN [25], and CMAN [28]. Due to the fact that these models need enough sampling data to train, 1%–5% data of the unobservable PV units are reserved for training, and the other data of these units are discarded for testing the imputation performance. In the experiments, the Transformer model uses the multihead attention mechanism (see (8) and (9) in [15]) to capture the long-term periodic characteristics of the time-series and only uses one decoder to impute the missing value via an end-to-end way. In MVSTGN, the graph $\mathcal{G}$ in Section III, preliminaries of [25], is the same as the definition, $\mathcal{G}$ of this article, and the missing value is imputed by the node features and the trend features extracted from the spatial and temporal views. In CMAN, to simplify the objective function (see (21) in [28]), only the relation extraction and the first-phase entity recognition are considered (see (24) and (25) in [28]), and the parameters σ_1 and σ_2 are set as 1.

In addition to the deep learning models, some GAN-based models are also selected, including SolarGAN [17] and C-StyleGAN2-SE [20]. Among them, SolarGAN is trained by an unsupervised way that uses the data of the observable PV units as input and uses the reserved 1%–5% data as output. The other discarded data are used for testing the performance. In contrast, C-StyleGAN2-SE uses the additional meteorological data as input, and then outputs the completely missing time-series data within the specific period.

To ensure the sufficient computational power, all the experiments are made on a server with 2.1-GHz Intel E5-2620 CPU, 256 GB of RAM, and NVIDIA P100 GPU.

D. Imputation Performance of CM-GAN

The imputation performances of CM-GAN and the other baseline models on the NC and AZ dataset are presented in Tables I and II, respectively. It is clear that the performances of CM-GAN on the two datasets are better than that of the other baseline models at different unobservability rates. That is, CM-GAN achieves the state-of-the-art imputation performance and performs stably on different datasets with different unobservability rates. Compared with C-StyleGAN2-SE, which is the second best GAN-based model, CM-GAN achieves 1.0%–7.3% improvement of RMSE and 1.3%–8.4% improvement of MAE on the NC dataset and achieves 18.4%–21.8% improvements of RMSE and 27.1%–30.7% of MAE on the AZ dataset.

The trajectories of the real data and the data imputed by each model when $\eta = 5\%$ are shown in Fig. 6. By visual inspection, the imputed data of CM-GAN almost coincide with the real data. The periodic characteristics in real data are detected, even with sudden changes in the PV power output. In contrast, the imputed data of other baseline models, including BGCP, LATC, MVSTGN, CMAN, and C-StyleGAN2-SE, deviate from the real data. For the SolarGAN and Transformer, their imputed data have sharp fluctuations.

E. Ablation Studies on Cross Modals

Some ablation studies are conducted to show the contribution of each modal. The experimental results are presented in Tables I and II, where “-G,” “-C,” and “-T” denote the CM-GAN only with graph generated from GPS data, capacity labels, and sequence tag as input, respectively. Some models have two labels, e.g., CM-GAN-G-C. CM-GAN-G-C has the GPS data and capacity labels as input together.

First, the sequence tag is dropped, and then the performance of CM-GAN declines dramatically whatever on the NC dataset or AZ dataset. The trajectory of the imputed data is shown in

TABLE II
IMPUTATION PERFORMANCE OF EACH MODEL ON THE AZ DATASET

η	5%		10%		15%		20%		25%	
	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE
BGCP	0.128	0.111	0.142	0.123	0.118	0.101	0.116	0.099	0.115	0.098
LATC	0.189	0.147	0.197	0.154	0.203	0.160	0.212	0.165	0.219	0.169
Transformer	0.121	0.097	0.124	0.101	0.126	0.103	0.127	0.105	0.128	0.105
MVSTGN	0.118	0.089	0.119	0.091	0.123	0.093	0.124	0.096	0.127	0.097
CMAN	0.094	0.070	0.095	0.073	0.098	0.074	0.099	0.074	0.101	0.077
SolarGAN	0.119	0.093	0.119	0.092	0.120	0.098	0.118	0.096	0.119	0.097
C-StyleGAN2-SE	0.101	0.083	0.103	0.084	0.103	0.085	0.104	0.088	0.105	0.089
CM-GAN	0.079	0.058	0.082	0.059	0.084	0.062	0.084	0.061	0.083	0.062
CM-GAN-G-C	0.263 (-70.0%)	0.211 (-72.5%)	0.263 (-68.8%)	0.212 (-72.2%)	0.264 (-68.2%)	0.213 (-70.9%)	0.265 (-68.3%)	0.214 (-71.5%)	0.265 (-68.7%)	0.214 (-71.0%)
CM-GAN-T	0.105 (-24.8%)	0.083 (-30.1%)	0.108 (-24.1%)	0.083 (-28.9%)	0.110 (-23.6%)	0.087 (-28.7%)	0.110 (-23.6%)	0.087 (-29.9%)	0.111 (-25.2%)	0.087 (-28.7%)
CM-GAN-T-G	0.085 (-7.1%)	0.065 (-10.8%)	0.085 (-3.5%)	0.066 (-10.6%)	0.087 (-3.4%)	0.068 (-8.8%)	0.089 (-5.6%)	0.070 (-12.9%)	0.089 (-6.7%)	0.071 (-12.7%)
CM-GAN-T-C	0.096 (-17.7%)	0.076 (-23.7%)	0.098 (-16.3%)	0.077 (-23.4%)	0.101 (-16.8%)	0.078 (-20.5%)	0.102 (-17.6%)	0.078 (-21.8%)	0.103 (-19.4%)	0.079 (-21.5%)

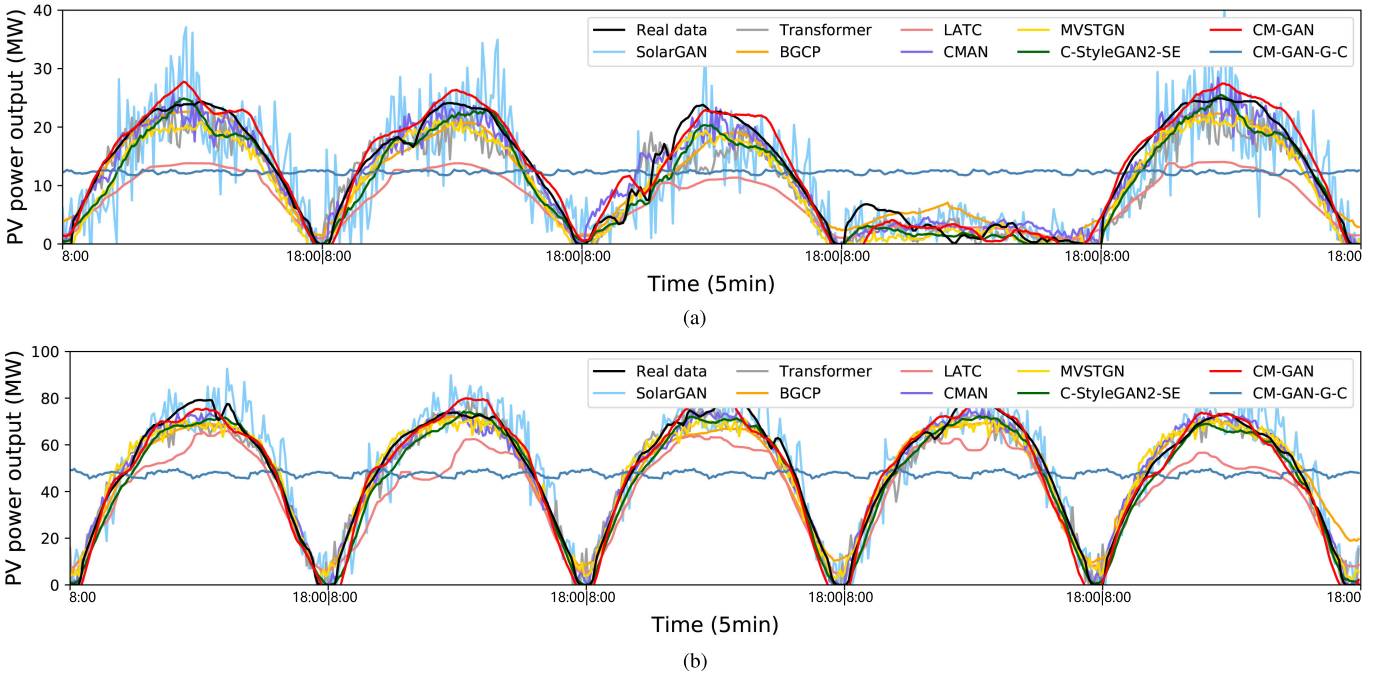


Fig. 6. Trajectories of the real data and the data imputed by each model when $\eta = 5\%$. (a) NC dataset. (b) AZ dataset.

Fig. 6. As is observed, the trajectory is a curve that cannot reflect the trend. Because the graph vector and the capacity label of each input are the same, and then the generated time-series also does not change. Thus, the sequence tag, namely, temporal modal, is necessary for CM-GAN because it records the trend information.

Then, the performance of CM-GAN only with temporal modal, namely, CM-GAN-T, is tested and recorded in the tables. The performance indices are lower than these of CM-GAN with 20.2%–22.9% decline on RMSE and 18.5%–23.5% decline on MAE. One possible reason is that CM-GAN needs graph vector and capacity label to know the power output range of the PV unit. Thus, the RMSEs and MAEs of CM-GAN-T tested on each PV unit are collected, and the

variances of the RMSEs and MAEs are presented in Fig. 7. Clearly, the variances gradually increase and stabilize finally as the unobservability rate increases. Thus, without graph vector and capacity label as input, CM-GAN-T cannot know the power output range of the PV unit. As the unobservability rate increases, the variances of RMSEs and MAEs gradually increase and stabilize finally because the numbers of capacity of the unobservable PV units are also gradually diversified and stabilized.

Accordingly, to further explain the contributions of graph vector and capacity label, the performances of CM-GAN-T-G and CM-GAN-T-C are also tested and recorded in the tables. Clearly, the two modal data provide information for imputation performance. As can be seen in Fig. 8, the time-series Pearson

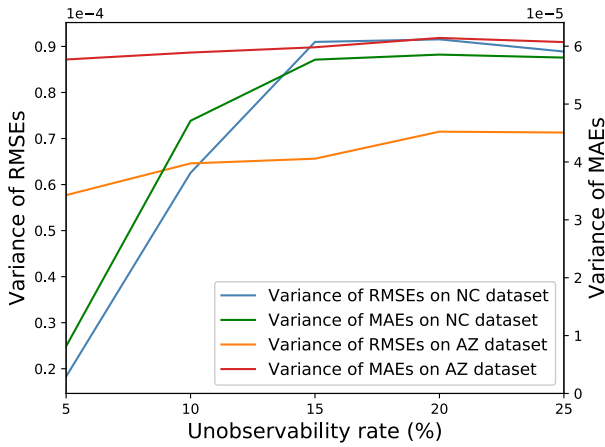


Fig. 7. Variance of RMSEs and MAEs becomes stabilized as η increases when only temporal modal holds.

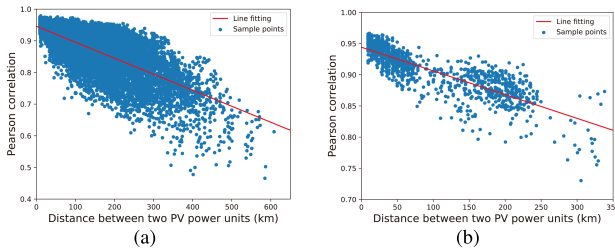


Fig. 8. Time-series Pearson correlation of two power units is negatively related to the distance of them. (a) NC dataset. (b) AZ dataset.

TABLE III
ABLATION STUDIES ON SETTINGS OF NEURAL UNITS

Neural units	NC dataset		AZ dataset	
	RMSE	MAE	RMSE	MAE
G : [256, 256, 256] D : [256, 256, 256]	0.109	0.081	0.083	0.063
G : [512, 512, 512] D : [512, 512, 512]	0.104	0.076	0.079	0.058
G : [768, 768, 768] D : [768, 768, 768]	0.103	0.076	0.079	0.058
G : [1024, 1024, 1024] D : [1024, 1024, 1024]	0.103	0.075	0.079	0.058
G : [1024, 1024, 1024] D : [512, 512, 512]	0.103	0.075	0.079	0.058

correlation of two power units is negatively related to the distance of them. Due to the fact that the relationship is strongly negative, especially on the AZ dataset, the completely missing time-series data can be recovered from the distance information. Thus, it is explainable and reasonable that the state-of-the-art imputation results of CM-GAN benefit from the cross-modal data fusion technique.

F. Ablation Studies on the Settings of CM-GAN

Some ablation studies have been conducted to select the appropriate settings of CM-GAN. At first, each hidden layer of the generator G and the discriminator D is assigned by different numbers of neural units, respectively. The results of the ablation studies are presented in Table III. As is observed, when the number of neural units is larger than 512, the performance of CM-GAN has almost no change. That is, the neural networks have enough ability to fit the data. Thus, the settings of the hidden layers in the proposed CM-GAN are reasonable.

TABLE IV
ABLATION STUDIES ON ACTIVATION FUNCTION

λ	NC dataset		AZ dataset	
	RMSE	MAE	RMSE	MAE
0	0.114	0.086	0.083	0.064
0.1	0.103	0.075	0.079	0.058
0.2	0.103	0.075	0.079	0.058
0.3	0.103	0.075	0.079	0.058
0.4	0.103	0.075	0.079	0.058
0.5	0.104	0.076	0.079	0.058

Then, the proposed CM-GAN uses Leaky ReLU to be the activation function (see Fig. 5). To select appropriate λ for Leaky ReLU function, the value of λ of each hidden layer is assigned by different numbers ranging from 0 to 0.5. The results of the ablation studies are presented in Table IV. When the value of λ is set to be 0, that is, standard ReLU function, the performance of CM-GAN declines. In contrast, if λ is set to be nonzero, the performance of CM-GAN has almost no change. Thus, it is reasonable to set λ as 0.2 in the proposed CM-GAN.

G. Improving Time-Series Predictability by CM-GAN

Another experiment is conducted to further verify the valid application scenario. In this article, it is verified that the generated PV power output data can be used to improve the accuracy of PV power output prediction models. During the experiment, the extra PV power output data are generated by CM-GAN, and then mixed into the remained real data, which means two training datasets are obtained, namely, real dataset and mixed dataset. The predictive accuracy of the designated model is tested on the real dataset and the mixed dataset, respectively, with the same parameter settings. Without loss of generality, the common LSTM model [35] is selected to complete the aforementioned process and provide a comparison of performance improvement. To ensure the validity and practicality, the model architecture and implement procedures of LSTM keep the same as the concerning paper [35].

The results of this experiments are all displayed in Table V. It shows the predictive accuracy of LSTM on different prediction steps, namely, 1 and 4 h. It is clear that the performances of LSTM trained on the mixed dataset are generally better than that only trained on the real dataset, which is concluded from the improvement value in brackets. Practically, the 4-h PV power output prediction remains an integral part which needs to be better solved. However, as the prediction steps increase, the predictive accuracy decreases because of the increasing difficulty of long-term data fitting. The 4-h results presented in Table V under different experimental conditions are obviously worse than the 1-h results, whereas the data generated by CM-GAN still can improve the results when the prediction steps reach 4 h. It is reason that the generated data of CM-GAN provide more predictability information for the prediction of LSTM. Thus, with the usage of mixed dataset generated by CM-GAN, the prediction accuracy of the LSTM-like deep learning methods can be further improved when the predictive object is long-term PV power output.

TABLE V
PREDICTION ACCURACY OF PV POWER OUTPUT PREDICTION USING LSTM MODEL WITH REAL AND MIXED DATASETS

η		5%		10%		15%		20%		25%	
		Real	Mixed	Real	Mixed	Real	Mixed	Real	Mixed	Real	Mixed
NC (1h)	RMSE	0.142	0.131(+7.7%)	0.136	0.127(+6.6%)	0.136	0.116(+14.7%)	0.135	0.124(+8.1%)	0.125	0.111(+11.2%)
	MAE	0.097	0.094(+3.1%)	0.092	0.089(+3.3%)	0.092	0.082(+10.9%)	0.094	0.088(+6.4%)	0.086	0.078(+9.3%)
NC (4h)	RMSE	0.176	0.171(+2.8%)	0.206	0.196(+4.9%)	0.177	0.182(-3.0%)	0.198	0.189(+4.5%)	0.184	0.166(+9.8%)
	MAE	0.134	0.125(+6.7%)	0.160	0.150(+6.3%)	0.137	0.139(-1.5%)	0.151	0.143(+5.3%)	0.144	0.126(+12.5%)
AZ (1h)	RMSE	0.117	0.112(+4.3%)	0.108	0.109(-0.9%)	0.111	0.104(+6.3%)	0.111	0.108(+2.7%)	0.110	0.106(+3.6%)
	MAE	0.075	0.074(+1.3%)	0.069	0.069(+0.0%)	0.071	0.069(+2.8%)	0.070	0.071(-1.4%)	0.072	0.069(+4.2%)
AZ (4h)	RMSE	0.127	0.105(+17.3%)	0.158	0.104(+34.2%)	0.157	0.152(+3.2%)	0.156	0.159(-2.0%)	0.171	0.124(+27.5%)
	MAE	0.085	0.068(+20.0%)	0.113	0.066(+41.6%)	0.108	0.108(+0.0%)	0.109	0.110(-0.9%)	0.122	0.083(+32.0%)

IV. CONCLUSION

In this article, CM-GAN is proposed to address the CDM problem widely existing in modern digital industry, especially in the PV power system. Different from other GAN-based models, CM-GAN combines the cross-modal data fusion technique and the deep adversarial generation technique to construct a cross-modal data generator. The cross-modal data fusion module is specially designed to fuse different modal data, and it enables GAN to generate time-series PV power output data from GPS data and capacity labels of each PV unit.

Lots of experiments are conducted to compare the performance of CM-GAN with that of BGCP, LATC, Transformer, MVSTGN, CMAN, SolarGAN, and C-StyleGAN2-SE. The experimental results show that CM-GAN achieves generally better performances on two datasets than the other baseline models. That is, CM-GAN achieves the state-of-the-art performance. Moreover, ablation studies have been conducted to verify the contribution of the cross-modal data fusion technique. Some other ablation studies have been conducted to explain the parameter settings of CM-GAN. Apart from this, prediction experiments are also designed as downstream tasks of CDM imputation. The experimental results show that the generated data of CM-GAN can provide more predictability information for improving predictive accuracy of deep learning method.

Our future work will focus on improving the performance of CM-GAN. The cross-modal mechanism is likely to be realized better by self-attention technique [36], which can simplify the encoding process from multiple modal data to target modal data. However, this does not affect the academic value about the insight that CM-GAN fills the gap of CDM problem. On the other hand, we will pay more attention on expanding renewable energy application of CM-GAN, including wind power, tidal energy [37], [38], etc. In different CDM scenarios, the way to extract modal features may also be different. We believe the CM-GAN framework will become more general for various CDM scenarios of future energy industry.

REFERENCES

- [1] C. Chen, S. Rosa, C. X. Lu, B. Wang, N. Trigoni, and A. Markham, "Learning selective sensor fusion for state estimation," *IEEE Trans. Neural Netw. Learn. Syst.*, early access, Jun. 3, 2022, doi: [10.1109/TNNLS.2022.3176677](https://doi.org/10.1109/TNNLS.2022.3176677).
- [2] T. Baltrušaitis, C. Ahuja, and L.-P. Morency, "Multimodal machine learning: A survey and taxonomy," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 41, no. 2, pp. 423–443, Feb. 2019.
- [3] Y. Liu, X. Zhang, F. Huang, L. Cheng, and Z. Li, "Adversarial learning with multi-modal attention for visual question answering," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 32, no. 9, pp. 3894–3908, Sep. 2021.
- [4] L. Zhen, P. Hu, X. Peng, R. S. M. Goh, and J. T. Zhou, "Deep multimodal transfer learning for cross-modal retrieval," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 2, pp. 798–810, Feb. 2022.
- [5] D. Wang, T. Zhao, W. Yu, N. V. Chawla, and M. Jiang, "Deep multimodal complementarity learning," *IEEE Trans. Neural Netw. Learn. Syst.*, early access, Apr. 18, 2022, doi: [10.1109/TNNLS.2022.3165180](https://doi.org/10.1109/TNNLS.2022.3165180).
- [6] C. Gu and P. Jirutitijaroen, "Dynamic state estimation under communication failure using Kriging based bus load forecasting," *IEEE Trans. Power Syst.*, vol. 30, no. 6, pp. 2831–2840, Nov. 2015.
- [7] M. Heidari Kapourchali, M. Sepehry, and V. Aravithan, "Fault detector and switch placement in cyber-enabled power distribution network," *IEEE Trans. Smart Grid*, vol. 9, no. 2, pp. 980–992, Mar. 2018.
- [8] X. Ge, Q. Han, X. Zhang, L. Ding, and F. Yang, "Distributed event-triggered estimation over sensor networks: A survey," *IEEE Trans. Cybern.*, vol. 50, no. 3, pp. 1306–1320, Mar. 2020.
- [9] X. Zhong, I. Jayawardene, G. K. Venayagamoorthy, and R. Brooks, "Denial of service attack on tie-line bias control in a power system with PV plant," *IEEE Trans. Emerg. Topics Comput. Intell.*, vol. 1, no. 5, pp. 375–390, Oct. 2017.
- [10] K. Wang, X. Qi, and H. Liu, "A comparison of day-ahead photovoltaic power forecasting models based on deep learning neural network," *Appl. Energy*, vol. 251, Oct. 2019, Art. no. 113315.
- [11] Q. Zhao, G. Zhou, L. Zhang, A. Cichocki, and S. Amari, "Bayesian robust tensor factorization for incomplete multiway data," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 27, no. 4, pp. 736–748, Apr. 2016.
- [12] X. Chen, Z. He, and L. Sun, "A Bayesian tensor decomposition approach for spatiotemporal traffic data imputation," *Transp. Res. C, Emerg. Technol.*, vol. 98, pp. 73–84, Jan. 2019.
- [13] M. Ko, K. Lee, J. Kim, C. W. Hong, Z. Y. Dong, and K. Hur, "Deep concatenated residual network with bidirectional LSTM for one-hour-ahead wind power forecasting," *IEEE Trans. Sustain. Energy*, vol. 12, no. 2, pp. 1321–1335, Apr. 2021.
- [14] M. Xia, H. Shao, X. Ma, and C. W. de Silva, "A stacked GRU-RNN-based approach for predicting renewable energy and electricity load for smart grid operation," *IEEE Trans. Ind. Informat.*, vol. 17, no. 10, pp. 7050–7059, Oct. 2021.
- [15] C. Wang, Y. Wang, Z. Ding, T. Zheng, J. Hu, and K. Zhang, "A transformer-based method of multienergy load forecasting in integrated energy system," *IEEE Trans. Smart Grid*, vol. 13, no. 4, pp. 2703–2714, Jul. 2022.
- [16] Y. Chen, Y. Wang, D. Kirschen, and B. Zhang, "Model-free renewable scenario generation using generative adversarial networks," *IEEE Trans. Power Syst.*, vol. 33, no. 3, pp. 3265–3275, May 2018.
- [17] W. Zhang, Y. Luo, Y. Zhang, and D. Srinivasan, "SolarGAN: Multivariate solar data imputation using generative adversarial network," *IEEE Trans. Sustain. Energy*, vol. 12, no. 1, pp. 743–746, Jan. 2021.
- [18] J. Qiao, T. Pu, and X. Wang, "Renewable scenario generation using controllable generative adversarial networks with transparent latent space," *CSEE J. Power Energy Syst.*, vol. 7, no. 1, pp. 66–77, Jan. 2021.

- [19] J. Liang and W. Tang, "Sequence generative adversarial networks for wind power scenario generation," *IEEE J. Sel. Areas Commun.*, vol. 38, no. 1, pp. 110–118, Jan. 2020.
- [20] R. Yuan, B. Wang, Y. Sun, X. Song, and J. Watada, "Conditional style-based generative adversarial networks for renewable scenario generation," *IEEE Trans. Power Syst.*, vol. 38, no. 2, pp. 1281–1296, Mar. 2023.
- [21] C. Zhang, Y. Cui, Z. Han, J. T. Zhou, H. Fu, and Q. Hu, "Deep partial multi-view learning," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 5, pp. 2402–2415, May 2022.
- [22] M. Yang, Y. Li, P. Hu, J. Bai, J. Lv, and X. Peng, "Robust multi-view clustering with incomplete information," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 1, pp. 1055–1069, Jan. 2023.
- [23] X. Peng, Z. Huang, J. Lv, H. Zhu, and J. T. Zhou, "Comic: Multi-view clustering without parameter selection," in *Proc. Int. Conf. Mach. Learn.*, 2019, pp. 5092–5101.
- [24] Y. Lin, Y. Gou, Z. Liu, B. Li, J. Lv, and X. Peng, "COMPLETER: Incomplete multi-view clustering via contrastive prediction," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 11169–11178.
- [25] Y. Yao, B. Gu, Z. Su, and M. Guizani, "MVSTGN: A multi-view spatial-temporal graph network for cellular traffic prediction," *IEEE Trans. Mobile Comput.*, vol. 22, no. 5, pp. 2837–2849, May 2023, doi: 10.1109/TMC.2021.3129796.
- [26] Y. Zeng, D. Cao, X. Wei, M. Liu, Z. Zhao, and Z. Qin, "Multi-modal relational graph for cross-modal video moment retrieval," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 2215–2224.
- [27] P. Zhang, Y. Li, Z. Huang, and X. Xu, "Aggregation-based graph convolutional hashing for unsupervised cross-modal retrieval," *IEEE Trans. Multimedia*, vol. 24, pp. 466–479, 2022.
- [28] S. Zhao, M. Hu, Z. Cai, and F. Liu, "Dynamic modeling cross-modal interactions in two-phase prediction for entity-relation extraction," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 34, no. 3, pp. 1122–1131, Mar. 2023.
- [29] A. Grover and J. Leskovec, "node2vec: Scalable feature learning for networks," in *Proc. 22nd ACM SIGKDD Int. Conf. Knowl. Discovery Data Mining*, Aug. 2016, pp. 855–864.
- [30] T. Mikolov, K. Chen, G. Corrado, and J. Dean, "Efficient estimation of word representations in vector space," 2013, *arXiv:1301.3781*.
- [31] M. Arjovsky, S. Chintala, and L. Bottou, "Wasserstein generative adversarial networks," in *Proc. Int. Conf. Mach. Learn.*, 2017, pp. 214–223.
- [32] M. Mirza and S. Osindero, "Conditional generative adversarial nets," 2014, *arXiv:1411.1784*.
- [33] (2021). *NREL*. [Online]. Available: <https://www.nrel.gov/grid/solar-power-data.html#panel2>
- [34] X. Chen, M. Lei, N. Saunier, and L. Sun, "Low-rank autoregressive tensor completion for spatiotemporal traffic data imputation," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 8, pp. 12301–12310, Aug. 2022.
- [35] M. Abdel-Nasser, K. Mahmoud, and M. Lehtonen, "Reliable solar irradiance forecasting approach based on choquet integral and deep LSTMs," *IEEE Trans. Ind. Informat.*, vol. 17, no. 3, pp. 1873–1881, Mar. 2021.
- [36] A. Vaswani et al., "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, 2017, pp. 1–15.
- [37] J. Li, J. Zhou, and B. Chen, "Review of wind power scenario generation methods for optimal operation of renewable energy systems," *Appl. Energy*, vol. 280, Dec. 2020, Art. no. 115992.
- [38] M. S. Chowdhury et al., "Current trends and prospects of tidal energy technology," *Environ., Develop. Sustainability*, vol. 23, no. 6, pp. 8179–8194, Jun. 2021.



Mingyu Kang received the master's degree in computer technology from Southeast University, Nanjing, China, in 2021. He is currently pursuing the Ph.D. degree with the Jiangsu Key Laboratory of Networked Collective Intelligence, School of Cyber Science and Engineering, Southeast University.

His research interests include complex system, complex network, deep learning, and causal inference.



Ran Zhu received the B.S. and M.S. degrees in electrical engineering from Nanjing Normal University, Nanjing, China, in 2012 and 2015, respectively. He is currently pursuing the Ph.D. degree in cyber science and engineering with Southeast University, Nanjing.

He was an Electrical Engineer with the State Grid Electric Power Research Institute (SGEPRI), Nanjing. His research interests include modeling, planning, and optimization of resilient power system.



Duxin Chen (Member, IEEE) received the B.S. degree in automatic control and the Ph.D. degree in control science and engineering from the Huazhong University of Science and Technology, Wuhan, China, in 2013 and 2018, respectively.

He is currently an Associate Professor with the Jiangsu Provincial Key Laboratory of Networked Collective Intelligence, School of Mathematics, Southeast University, Nanjing, China. His research interests include complex networks and systems, artificial-intelligence-related theory and applications.



Xiaolu Liu received the B.S. degree in automatic control from Southwest Petroleum University, Chengdu, China, in 2014, and the Ph.D. degree in control science and engineering from the Huazhong University of Science and Technology, Wuhan, China, in 2018, respectively. She is currently a Lecturer with the School of Automation, Nanjing Institute of Technology, Nanjing, China. Her research interests include multiagent systems, and complex networks and systems.



Wenwu Yu (Senior Member, IEEE) received the B.Sc. degree in information and computing science and the M.Sc. degree in applied mathematics from the Department of Mathematics, Southeast University, Nanjing, China, in 2004 and 2007, respectively, and the Ph.D. degree from the Department of Electronic Engineering, City University of Hong Kong, Hong Kong, China, in 2010.

He is currently the Founding Director of the Laboratory of Cooperative Control of Complex Systems, the Deputy Associate Director of Jiangsu Provincial

Key Laboratory of Networked Collective Intelligence, an Associate Dean with the School of Mathematics, and a Full Professor with the Endowed Chair Honor, Southeast University. He held several visiting positions in Australia, China, Germany, Italy, The Netherlands, and the USA. He has published about 100 IEEE TRANSACTIONS journal articles with more than 20 000 citations. His research interests include multiagent systems, complex networks and systems, disturbance control, distributed optimization, machine learning, game theory, cyberspace security, smart grids, intelligent transportation systems, and big-data analysis.

Dr. Yu is the recipient of the Second Prize of State Natural Science Award of China in 2016. He is also the Cheung Kong Scholars Programmer of Ministry of Education of China (Artificial Intelligence). He was listed by Clarivate Analytics/Thomson Reuters Highly Cited Researchers in Engineering from 2014 to 2022. He serves as an Editorial Board Member of several flag journals, including IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS-PART II: EXPRESS BRIEFS, IEEE TRANSACTIONS ON INDUSTRIAL INFORMATICS, IEEE TRANSACTIONS ON SYSTEMS, MAN AND CYBERNETICS: SYSTEMS, IEEE TRANSACTIONS ON INDUSTRIAL CYBER-PHYSICAL SYSTEMS, *Science China Information Sciences*, and *Science China Technological Sciences*.